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DIGEST

Sex-Selective Abortions During Past Three Decades May Explain Absence of Millions of Girls in India

AUTHORS

Lisa Melhado, Guttmacher Institute

Sex ratio data suggest that selective abortion of females increased substantially in India between 1980 and 2010, particularly among women whose only child was a daughter, according to a new analysis.[1] The sex ratio for second births among women with a firstborn daughter fell from 906 females per 1,000 males in 1990 to 836 females per 1,000 males in 2005; in contrast, no declines occurred in the sex ratio for firstborns or for second births after a firstborn son. Declines in the sex ratio were greater among mothers with 10 or more years of education than among those without any education, and greater among those from wealthier households than among those from poorer ones. From these and other data, the researchers estimate that the number of sex-selective abortions per decade rose from about one million in the 1980s to 2.6 million in the 1990s and 4.5 million in the 2000s, for a total of more than eight million.

Although the natural sex ratio—roughly 950–975 females per 1,000 males at birth in most high-income countries—generally varies little by birth order or by the sex of the previously born child, some evidence suggests that this is not the case in India. Moreover, the 2011 Indian census counted 7.1 million fewer girls than boys aged 0–6 years. Such findings, combined with persistent son preference and the increased availability of fetal ultrasound, led researchers to hypothesize that widespread selective abortion of females has affected sex ratios.

To assess trends in sex ratio by birth order between 1990 and 2005, and to determine if the ratio varies by women's wealth or education, researchers used data on more than 250,000 births from three rounds of the National Family Health Survey, a nationally representative survey of Indian households. Women of childbearing age were asked to provide a complete birth history (including each child's date of birth, birth order, gender and survival status), as well as information on their educational attainment, religion and household assets; the last were used to classify women into state-specific wealth quintiles. Differences in sex ratio trends by birth order and women's characteristics were identified using linear regression. To estimate the numbers of "missing" girls, the researchers compared the expected number of girls aged 0–6 (calculated from the natural sex ratio at birth and adjusted for age- and sex-

specific mortality) with the actual numbers enumerated in the 1991, 2001 and 2011 Indian censuses; they calculated low, high and mean estimates.

Between 1990 and 2005, the sex ratio for second births after a firstborn daughter declined by 0.5% annually, from 906 females per 1,000 males in 1990 to 861 in 1997 and 836 in 2005. During the same period, the sex ratios for firstborns and for second births after a firstborn son did not change, remaining close to the natural sex ratio. The sex ratio for second births after a firstborn daughter was unchanged among mothers with no education and among those in the lowest wealth quintile, but fell among mothers reporting 10 or more years of education and among those in the highest wealth quintile. Among women with two daughters but no sons, the sex ratio for third births fell from 882 in 1990 to 768 in 2005, though the decline did not reach statistical significance.

The researchers noted that if no selective abortion of females had occurred and the two sexes had had equal mortality rates, the imbalances in the sex ratio should have been smaller than those they encountered. They concluded that most of the differences in the sex ratio could be explained by sex-selective abortions. After adjustments for age- and sex-specific mortality (e.g., mortality is higher among girls than among boys for most of the first five years of life), the estimated number of selective abortions of females per decade rose from about one million in the 1980s (range of estimates, 0–2.0 million) to 2.6 million in the 1990s (1.2–4.1 million) and 4.5 million in the 2000s (3.1–6.0 million), yielding a total of 8.1 million selective abortions (4.2–12.1 million). At 2010 birth and child mortality rates, a 1% drop in the sex ratio among children aged 0–6 is consistent with an additional 1.2–3.6 million abortions of ?females.

Although sex ratios of fewer than 950 females per 1,000 males have become widespread in India—the number of districts with such ratios increased from 336 to 433 between 2001 and 2011—the rate of increase in selective abortions of females appears to be slowing; the rate increased by 260% between 1991 and 2001, and by 170% between 2001 and 2011.

The researchers acknowledge several limitations of their analysis: Year-to-year differences in sex ratios may have been due in part to random variation, the estimates of the number of selective abortions were necessarily crude, and the estimated biological sex ratio of 950–975 females per 1,000 males is based on European and North American populations and may not be applicable to India. Despite these limitations, the researchers conclude that selective abortion of females has increased in the past 30 years and has contributed to the widening gap in the child sex ratio. They suggest that the increases in sex-selective abortion may be due to son preference combined with declines in fertility; the finding that selective abortion of females appears to be especially common among educated and wealthy women may reflect that these women are better able than others to afford ultrasound and abortion services. The researchers note that a 1996 law aimed at curbing sex-selective abortion likely has been ineffective because few providers have been prosecuted; however, efforts to prevent selective abortions of females could benefit from "reliable monitoring and reporting of sex ratios by birth order in each of India's districts."—*L. Melhado*

REFERENCE

1. Jha P et al., Trends in selective abortions of girls in India: analysis of nationally representative birth histories from 1990 to 2005 and census data from 1991 to 2011, *Lancet,* 2011, 377(9781):1921–1928.

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